

Martin Bissonnette

Nanotechnology Engineering Student



343-370-3399



LinkedIn



m2bisson@uwaterloo.ca



GitHub



ResearchGate



Portfolio

Skills

Lab Equipment AFM, SEM, XRD, Microscopy, Spectroscopy (UV-Vis, FTIR, Raman), TMA, TGA, DSC

Lab Experience Bio/Organic Chemistry Techniques, Cleanroom Protocols, Optical Alignment, Laser Operation & Safety

Software SolidWorks, AutoCAD, Fusion, FEMM, KLayout, KiCAD, Gwyddion, ImageJ/Fiji

Programming Languages Python, MATLAB, Excel VBA, C++, HTML, CSS

Languages Fluent in English and French (DELFI Certified)

Work Experience

Irradiant Technologies

September 2025 – May 2026

Research Engineer

Boston, Massachusetts

- Developed novel semiconductor packaging methods for proprietary nano-porous scaffold materials, increasing yield and throughput by 10x and allowing the company to acquire a \$500K contract with Coherent.
- Used AFM and SEM systems at Harvard CNS to validate samples and meet DARPA Additive Manufacturing of Microsystem (AMME) milestones, using Gwyddion and Fiji for accurate data analysis.
- Automated chemical experimentation by integrating an Opentrons OT-2 platform, including custom labware design and a user-facing control interface.
- Designed and built an optical system to measure insertion losses of photonic waveguides to find ideal geometries and print conditions.
- Contributed to the design and optimization of corrective optics for photonic interconnects specialized for data centers for companies such as Nvidia, Coherent and Corning.
- Worked on process optimization for stitching nano-fab structures together using DLW systems to create conductive centimeter-scale structures with nanoscale resolution.
- Co-authored a low-cost open-access two-photon lithography system for academic research, published on ResearchGate. ([link](#))

C2MI

January 2025 – May 2025

Assembly Process Intern

Bromont, Quebec

- Performed microfabrication and testing in a Class 1000 cleanroom using semiconductor and optical packaging tools.
- Performed R&D on photonic wire bonds as an alternative photonic interconnect method in collaboration with IBM.
- Created a procedure to create laser-induced graphene patterns for electroplating as a novel method of PCB creation.
- Documented and tested a wafer-level packaging method to embed chips into remolded wafers, allowing further processing and higher package density.
- Characterized materials and assembly quality using microscopy, spectroscopy, and electro-mechanical analysis techniques.

SMART LAB

May 2025 – August 2025

Part Time Research Assistant

Waterloo, Ontario

- Synthesized hydrogels according to lab procedure using sterile wet lab techniques.
- Conducted quantitative material characterization to evaluate and tailor hydrogel properties for targeted drug-delivery.

Lactalis Canada

2024 Summer Co-op

Engineering Student

Ingleside, Ontario

- Developed a data-analysis program for sensor data, optimizing heat exchanger placement and improving energy efficiency.
- Created CAD designs for parts and comprehensive floor plans, ensuring precise planning and streamlined manufacturing processes.
- Analyzed machine processing rates to identify production bottlenecks, quantifying cost savings and enhancing overall throughput.
- Optimized equipment safeguard designs to ensure regulatory compliance and worker safety.

Projects

UW Silica Integrated Optics Co-founder

December 2023 – Ongoing

- Established an official University design team specifically aimed at Nanotechnology Engineering students.
- Led research and fabrication of a high-speed programmable laser shutter system using Arduino and hardware components.
- Actively work in the optics lab creating DOEs to test for optimal laser parameters for desired voxel creation.

Education

University of Waterloo

Fall 2023 – Spring 2028

Bachelor of Applied Sciences in Honours Nanotechnology Engineering – 82% Avg.

Ontario, Canada